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Satellite Trail Estimates for Astrophotography with the 12" Irving Porter Church Memorial Refractor

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ABSTRACT

J. Rayworth et al. (2026) estimate the incidence of artificial satellite–induced trails in Vera C. Rubin Observatory images at ∼0.34 trails image^{-1}. Following their analysis and using the typical astrophotography setup of the Cornell Astronomical Society circa 2026 with the Irving Porter Church Memorial Refractor Telescope, we estimate the incidence of satellite trails in raw ("light") frames to be as high as 0.95 trails image^{-1} in an uncropped 3-minute exposure, but more like 0.60 trails image^{-1} if one employs an 80% crop in both height and width to remove edge effects from the image. We show that the optimum S/N for a stacked image is achieved using 154-second exposures for images cropped in the usual way.

1. INTRODUCTION

The Irving Porter Church Memorial Refractor Telescope ("Irv") was installed at Cornell University's Fuertes Observatory on October 16, 1922. Astrophotography with Irv began shortly after the telescope's installation, but the Cornell Astronomical Society (CAS; founded in 1972) has used its current camera for only the past ∼3 yr. A typical CAS astrophotography run uses ∼1–2 hr of total exposure time, during which individual exposure intervals range from ∼10 s to ∼3 min. Longer exposures have been useful for, e.g., capturing detail in the spiral arms of nearby galaxies, but with an upper limit of ∼20–40 such exposures per session, the necessity of removing any exposures with satellite trails often dramatically reduces the amount of total viable integration time, with unfortunate consequences for the S/N of the final image. In this note, we aim to use the analysis of J. Rayworth et al. (2026) to estimate the satellite trail rate of incidence in images using CAS's current astrophotography setup with Irv and make recommendations for future sessions.



Figure 1. A contrast-boosted uncropped light frame of the galaxy M51 next to a Stellarium screenshot showing the same FoV. Using the declinations of the Gaia DR3 stars 1551999459048397568 and 1551892291023590400, we estimate the height of the frame at ∼27′ and, using the 3:2 aspect ratio, the width at ∼18′. The FoV is 0.135 deg^2.

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1.1. *CAS–Irv Astrophotography Setup*

Irv is a 12" f/15 refractor with a 2" eyepiece focuser. CAS typically uses a Canon EOS 6D Mk II DSLR camera, attached to the focuser using a conventional adapter.

The left panel of Figure 1 shows a typical uncropped light frame from this setup. (The frame has had a minimal contrast boost applied but is otherwise unprocessed.) The right panel shows the same field of view (FoV) as seen in the sky atlas program Stellarium. We use the declinations of two Gaia DR3 stars near the edges of the frame (see figure caption) to estimate the FoV solid angle as 0.135 deg^2.

In this calculation, we neglect the low-sensitivity corners of the frame, which see very little light due to blockage by Irv's architecture. Typical CAS astrophotographs are cropped to various sizes and aspect ratios, so our FoV estimate should be taken as an approximation.

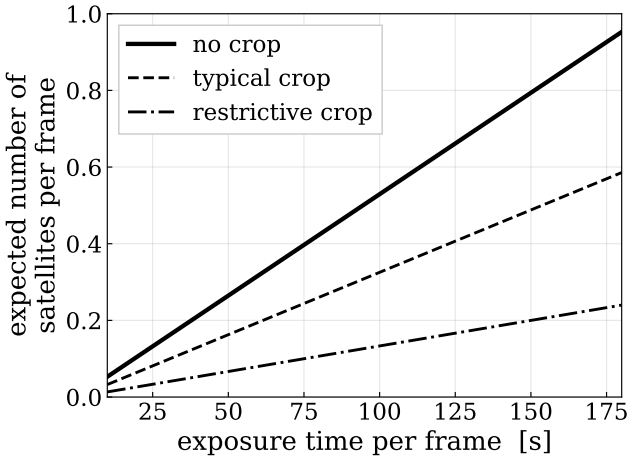


Figure 2. The expected number of satellites in a single frame as a function of the frame's exposure time for three representative FOVs. Exposure times shown are in the range [10 s, 180 s], with 10 s being typical for bright open clusters and 180 s being typical for galaxies. The "typical crop" case is 5000 pixels × 3200 pixels, or roughly 80% of the original FOV in both width and height. The "restrictive crop" case, designed to limit the FOV to the region immediately surrounding many deep-sky targets with minimal background star field, is 50% of the original FOV in both width and height.

2. SATELLITE TRAIL RATE

J. Rayworth et al. (2026) report the plane-of-sky satellite flux as $(3.51 \pm 1.5) \times 10^{-3}$ satellites deg^{-2} min^{-1} for satellites above +6.15 mag. They derive an empirical power law for extending this flux to dimmer magnitudes:

$$N(M_2) = N(M_1) r^{M_2 - M_1}, \tag{1}$$

where

$$r = \begin{cases} 4.62 & M \leq 6.15 \\ 1.72 & M > 6.15. \end{cases} \tag{2}$$

Since satellites are not stationary point sources but have their power smeared over several pixels, it remains to find the effective per-pixel magnitude of a trail. J. Rayworth et al. (2026) do this by estimating the number of pixels over which a satellite trail passes,

$$N_{\rm pix} = \frac{\omega_{\rm sat} t_{\rm exp}}{\theta_{\rm pix}}, \tag{3}$$

where $\omega_{\rm sat}$ is the angular speed of the satellite, $t_{\rm exp}$ is the exposure time, and $\theta_{\rm pix}$ is the angular size of a pixel. J. Rayworth et al. (2026) use $\omega_{\rm sat} \approx 0.56\,{\rm deg\,s^{-1}}$ in their calculations. Due to our camera's small FoV, we assume that all satellite trails will cross the full field (rather than originating or ending partway), so $t_{\rm exp} = t_{\rm pass}$ in this case, where $t_{\rm pass}$ is the average passage time, so Equation 3 reduces to

$$N_{\rm pix} \approx \frac{\theta_{\rm pass}}{\theta_{\rm pix}}, \tag{4}$$

where $\theta_{\rm pass}$ is the typical angular size of a satellite trail. Using a numerical simulation, J. Rayworth et al. (2026) report that

$$\theta_{\rm pass} \approx \frac{\sqrt{({\rm width_{FOV}})^2 + ({\rm height_{FOV}})^2}}{2}, \tag{5}$$

i.e., half the FOV diagonal, is accurate to within 10%. This value is $\theta_{\rm pass} \approx 0.27$ deg for the typical CAS setup.

The native size of a frame from CAS's Canon EOS 6D Mk II DSLR camera is 20.8 in × 13.867 in at a resolution of 300 pixels in^{-1}. Our estimate of 27′ for the angular size of the long dimension in Figure 1 gives us

$$\theta_{\rm pix} \approx \frac{27}{20.8 \times 300} \approx 0.0043', \tag{6}$$

or 7.2×10^{-5} deg. Thus,

$$N_{\rm pix} \approx \frac{0.27}{7.2 \times 10^{-5}} \approx 3744. \tag{7}$$

The magnitude loss (Equation 15 of J. Rayworth et al. 2026) is

$$\Delta m = 2.5 \log \left(\frac{N_{\rm pix}}{\rm FWHM} \right), \tag{8}$$

where FWHM is the full width at half maximum of the system's point spread function. Using the M51 image shown in Figure 1 at the original resolution, we estimate a FWHM of ∼10 pixels, giving us a magnitude loss of $\Delta m \approx 6.4$. Using +17 mag as Irv's approximate limiting magnitude for a stationary point source, we obtain a limiting magnitude for a typical satellite trail of $17 - 6.4 = 10.6$ mag.

For $M_1 = 6.15$ and $M_2 = 10.6$, we use Equation 1 to find that the plane-of-sky satellite flux brighter than Irv's limiting satellite magnitude is $(3.51 \times 10^{-3}) \times 1.72^{10.6-6.15} \approx 0.039$ satellites deg^{-2} min^{-1}.

Figure 2 shows the expected number of satellites per frame as a function of exposure time, assuming a satellite flux of 0.039 satellites deg^{-2} min^{-1} and an FoV of 0.135 deg^2 (black line). After cropping to remove the low-sensitivity corners, a typical FoV is more like 5000 pixels × 3200 pixels, or 0.083 deg^2. A more extreme crop that may nonetheless be suitable for many deep-sky objects with low angular sizes[2] is to take the central half

[2] For instance, M27 (Dumbbell Nebula) or M51 (Whirlpool Galaxy), but not M31 (Andromeda Galaxy) or M42 (Orion Nebula).

of the FOV in both width and height, which results in an FOV of $0.135/4 \approx 0.034\,\mathrm{deg}^2$. The latter two cases are also shown in Figure 2.

3. OPTIMUM EXPOSURE TIME

From the Central Limit Theorem, stacking N exposures reduces the noise in the final image by a factor of $\sqrt{N}$ over the single-frame case. This also holds true for a single exposure integrated for varying lengths of time (c.f. the radiometer equation in radio astronomy; Ch. 3 in J. J. Condon & S. M. Ransom 2016); that is, a 120-second exposure has a factor of $\sqrt{2}$ lower root-mean-square (rms) noise than a 60-second exposure. In fact, for an ideal astrophotography setup,[3] assuming a finite amount of time in one's session, there is no difference between taking a single long exposure and taking many short exposures.

However, read noise is a limiting factor at short exposure times, motivating exposure times above 10 s (and ideally above 20–30 s) with the CAS setup. The high-exposure-time end is limited by Irv's imperfect sidereal tracking and the likelihood that an exposure is disrupted by satellite trails or footsteps in the dome. The upper limit for CAS's setup has historically been set at 3 min. In this section, we derive an optimum exposure time within [10 s, 180 s] such that the S/N is maximized with respect to both read noise (at the short-exposure end) and frame discarding (at the long-exposure end).

We will neglect the effects of shot noise here because the total integration time is constant.[4] We will also assume that fixed-pattern noise in light frames is well mitigated by flat frames and that thermal noise is well mitigated by dark frames. The offset due to the read noise floor is mitigated by bias frames.

The calibrated and stacked image then consists of a signal S, built up from photons originating from the imaged source, and noise N, which has many sources but is dominated for a well-calibrated frame by shot noise (which we are ignoring here because it has the same contribution regardless of exposure time) and read noise.

The signal contribution increases linearly with total integration time $T_{\mathrm{tot}} = N_{\mathrm{frames}} \times t_{\mathrm{exp}}$. The read noise contribution increases as $\sqrt{N_{\mathrm{frames}}}$. The S/N therefore

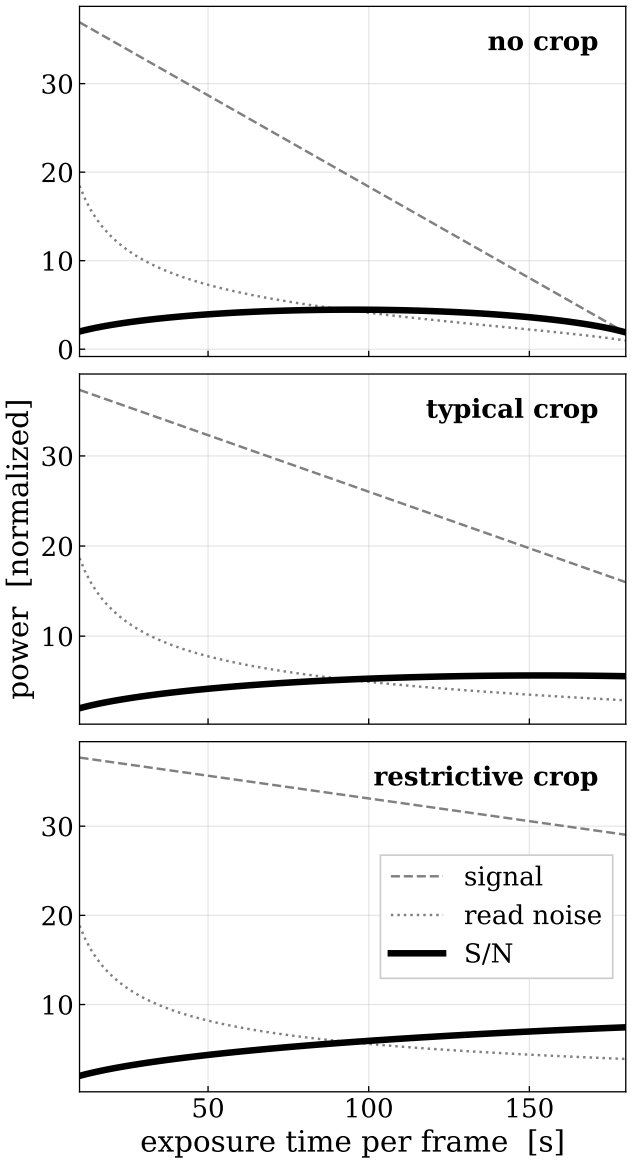


Figure 3. Total S/N (i.e., of the final stacked image) versus single-frame exposure time, shown for the same three cases as Figure 2. The individual components S (signal) and N (noise) for each case are also shown (N is the same across all three cases). The signal decreases with single-frame exposure time because more frames are discarded due to satellite trails, though this effect is mitigated by cropping. The noise contribution is highest at low exposure times but falls off as $\sqrt{N_{\mathrm{frames}}}$. The exposure time corresponding to maximum S/N is 94 s for the uncropped case, 154 s for the typical crop, and 375 s (beyond the limit of the x-axis shown) for the restrictive crop.

[3] No read noise or shot noise (R. S. Wright 2018a), perfectly dark sky, perfect sidereal tracking, etc.

[4] Shot noise is variability in the number of photons striking the camera from one time unit to the next. Photon arrival times can be modeled as a Poisson process. One can show that shot noise increases as the square root of the signal, so it is reduced equally well by taking longer exposures or by taking more exposures, so long as the total integration time is the same.

follows the scaling

$$\mathrm{S/N} \sim t_{\mathrm{exp}} \sqrt{N_{\mathrm{frames}}}, \tag{9}$$

where we have absorbed some constants having to do with the target's surface brightness, the camera gain, and the read noise into the "∼".

Figure 3 shows the scaling of Equation 9 for the same three cases as in Figure 2. The curves are normalized in all three cases so that S/N = 2 when $t_{\mathrm{exp}} = 10\,\mathrm{s}$. The optimum exposure time is 94 s for the uncropped case, 154 s for the typical crop, and 375 s for the restrictive crop. (At longer exposure times, sidereal drift may become a dominant noise source, so this is not necessarily to suggest that exposures of 375 s be employed with the current CAS setup.

Importantly, while S/N increases with total integration time, the location of the maximum (that is, the optimum t_{exp}) does not. Whether the observer has 1 hr or 10 hr to spend taking light frames, the optimum exposure time per frame is the same.

4. OTHER NOTES

The satellite rate computed in Section 2 assumes that the astrophotography session occurs during the twilight interval during which the observer is in darkness but objects in orbit may still be sunlit. This window is of order 1–3 hr (J. Rayworth et al. 2026). One possibility for satellite trail mitigation is to restrict long-exposure astrophotography campaigns to hours near midnight, though this is often impractical for student schedules.

This document is dated on the first page and may be revised as needed.

ACKNOWLEDGMENTS

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